AI Powered Intelligent Hospital Patient Flow Management System

CONCLUSION, FUTURE SCOPE AND REFERENCES

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1. Conclusion

CareFlow demonstrates an integrated, role-based hospital patient-flow prototype that connects registration, AI-assisted prioritisation, token queues, consultation, laboratory work, bed/ICU state and operational dashboards. The implementation is functional on the supplied local environment and all measured read endpoints passed the defined smoke run.

The strongest result is integration rather than clinical prediction. The wait regressor achieved R-squared 0.951041 on a generator-matched synthetic test and the triage classifier achieved 96.80% accuracy, but the latter recalled only 63% of critical cases. These results justify continued engineering investigation, not autonomous care decisions.

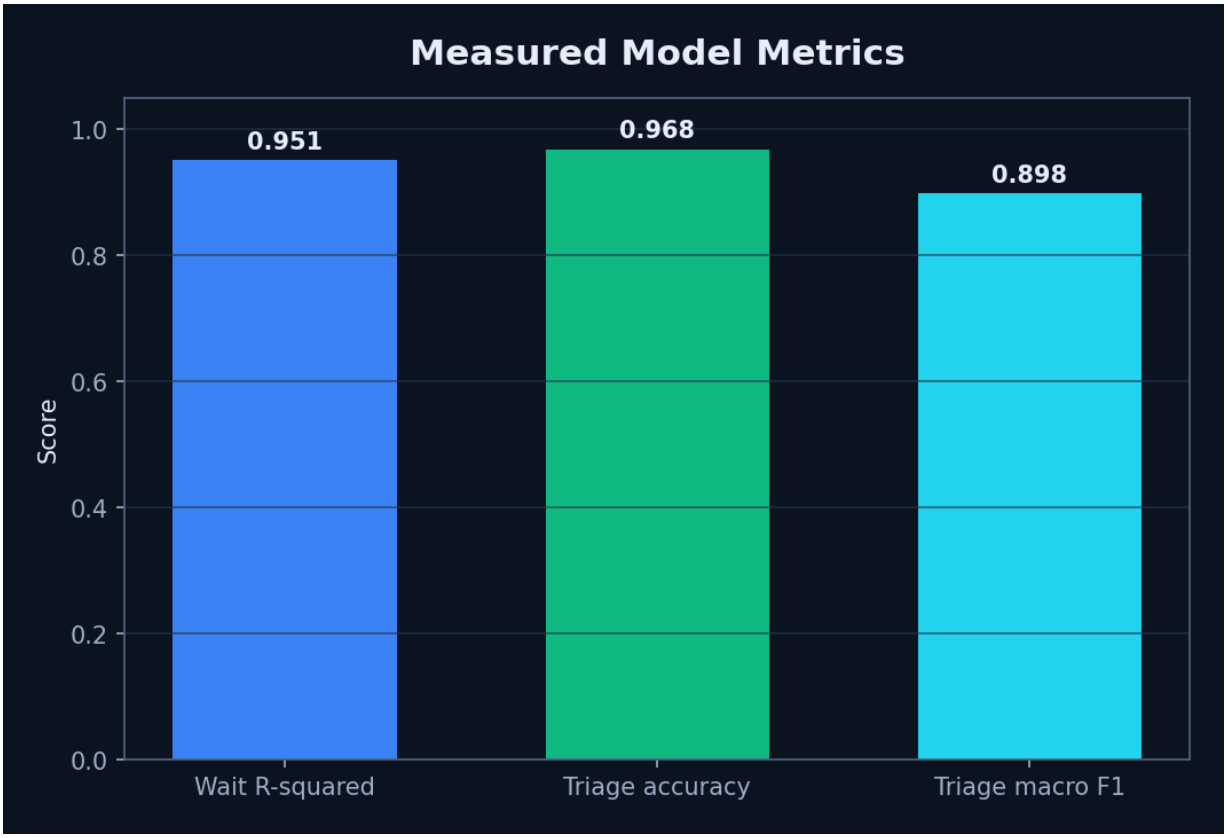


Figure 1. Summary of measured model results.

2. Key Findings

A single persisted Visit model lets role dashboards agree on token, status, priority and predicted wait. Priority-aware queue sorting is simple and auditable. Movement logs make explicit department changes inspectable. Bed and laboratory modules expose the operational dependencies that ordinary appointment applications omit.

The evidence also shows where design work remains: model safety is class-dependent, state transitions require tighter coupling, the initial nurse tab is faulty, and production security/scaling mechanisms are not present.

3. Future Scope

Table 1. Prioritised future scope.

Priority	Future work	Expected evidence
1	Validate on de-identified real patient-flow data with ethics and governance approval	External temporal test and subgroup metrics
2	Optimise critical recall with cost-sensitive learning and calibrated thresholds	Safety-focused confusion matrices
3	Integrate HL7 FHIR with hospital EHR, lab and admission systems	Interoperability conformance tests
4	Replace SQLite and in-process WebSocket with production services	Concurrent load, resilience and recovery tests
5	Authenticate live channels; rotate secrets; add audit and privacy controls	Security assessment
6	Fix dashboard state coupling and nurse default tab	Automated end-to-end tests
7	Forecast demand and recommend staffing/bed plans	Prospective operational study

4. Known Limitations

Both model datasets are synthetic and generated from designed distributions; they are not real patient records.

The measured model test partitions come from the same deterministic generators as training and are not independent clinical validation.

No real hospital deployment, clinician study, user-acceptance study, security audit, accessibility audit, or concurrent-load test was performed.

The 96.80% triage accuracy must be interpreted with the 63% critical-class recall; accuracy alone overstates safety-critical performance.

SQLite, a development JWT secret fallback, an unauthenticated WebSocket, and in-process broadcasting are unsuitable for production without hardening.

The landing-page claims of 42% wait reduction, 214 patients, and six departments are hard-coded marketing copy and are excluded from results.

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